

# Alexandra McDaniel

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## PERSONAL PROFILE

Graduate student in electrical engineering with a strong sense of initiative, a collaborative spirit, and enthusiasm for scientific discovery. Background research in power quality analysis and underwater acoustics. Looking to begin my career in the next 9 months.

## EDUCATION

**Master's in Electrical Engineering** | Laramie, WY Expected Graduation: May 2026  
*University of Wyoming*  
GPA 4.0

**Bachelor's in Applied Physics** | Provo, UT July 2024  
*Brigham Young University*  
GPA 3.75

**Rancho High School** | Las Vegas, NV June 2018  
*Clark County School District*  
GPA 4.8 (weighted) | Valedictorian

## WORK EXPERIENCE

**Graduate Research Assistant** | Laramie, WY August 2024 – Present  
*University of Wyoming Dept. of Electrical Engineering*

- Analyzed power quality disturbances and the effect they have on a transmission system with renewable energy connections

**Independent Research Contractor** | Remote August 2023 – Present  
*Brigham Young University Hydroacoustic Lab*

- Trained deep learning models for seabed classification using sound signatures from merchant ships

*Knobles Scientific and Analysis*

- Found evidence of deep sediment heterogeneity in the New England Mudpatch by performing statistical inference using 187 merchant ship noise datasets

*\* Funded by the Office of Naval Research*

**Field Technician** | Laramie, WY May – August 2024  
*University of Wyoming Dept. of Botany*

- Assisted with field surveys of vegetation and water potential in plants while leading the in-lab preparation and calibration of electrical field equipment

**Undergraduate Research Assistant** | Provo, UT January 2022 – July 2023  
*Brigham Young University Dept. of Physics and Astronomy*

- Characterized a laboratory water tank for sound measurements with variability in the sound speed

**Physics Lab Manager** | Provo, UT January 2022 – December 2022  
*Brigham Young University Dept. of Physics and Astronomy*

- Managed a team of six undergraduates with set-up of equipment and lab development for undergraduate level physics labs for seven courses

**Physics Lab Technician** | Provo, UT January – September 2019, May– December 2021  
*Brigham Young University Dept. of Physics and Astronomy*

- Led the re-design of two introductory labs for undergraduate students

## SKILLS

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- Coding languages:
  - Proficient: Python, MATLAB
  - Intermediate: MATLAB Simulink, LabVIEW, Mathematica
  - Beginner: C++, CRBasic, R
- Software:
  - Proficient: RT-Lab
  - Beginner: SolidWorks, Autodesk Inventor
- Management Experience:
  - Implementation of SCRUM and AGILE methodologies
  - Scheduling experience
  - Interviewing/hiring experience
  - Implementation of Trello and ClickUp
- Bilingual: English / Spanish

## PUBLICATIONS

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- Hopps-McDaniel, Alexandra M., et al. "Deep sediment heterogeneity inferred using very low-frequency features from merchant ships." The Journal of the Acoustical Society of America 156.4 (2024): 2265-2274.
- Hopps-McDaniel, Alexandra M., and Tracianne B. Neilsen. "Temperature-induced sound speed variability in a laboratory water tank." Proceedings of Meetings on Acoustics. Vol. 51. No. 1. AIP Publishing, 2023.
- Harmer, Madeline, et al. "Leveraging scientific modeling to engage pre-med undergraduates in physics lab courses." Physical Review Physics Education Research 20.2 (2024): 020150.
- Undergraduate Capstone Project: Hopps, Alexandra, "Characterizing the sound speed variability in laboratory water tank," Advisor: Traci Neilsen (Capstone, Jun 2023).

## PRESENTATIONS AT CONFERENCES

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- May 2025 - Acoustical Society of America (New Orleans, Louisiana) | "The impact of water sound speed on deep learning seabed classification from shipping noise"
- May 2024 - Acoustical Society of America (Ottawa, Canada) | "Deep sediment characterization from very low-frequency features from merchant ships"
- May 2023 - Acoustical Society of America (Chicago, Illinois) | "Introducing sound speed variability in a laboratory water tank"
- July 2022 - American Association for Physics Teachers (Grand Rapids, Michigan) | "Physics labs that resonate with pre-medicine students"

## RELEVANT COURSE WORK

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- Engineering: Statistical Signal Processing, Power Reliability, Power Quality, Power Engineering, Communication Theory, Cyber Physical Systems, Thermodynamics, Biomedical Engineering
- Mathematics: Linear Algebra, Calculus 3, Ordinary Differential Equations, Partial Differential Equations, Theory of Statistics 1 & 2
- Programming: Convolutional Neural Networks, Python lab, MATLAB lab, C++
- Physics: Electricity and Magnetism 1 & 2, Acoustics, Mechanics, Optics, Modern Physics, Intro to Physics
- TA courses: Basic Circuits Lab, Communication Theory Lab, Introductory Physics Lab

## VOLUNTEER EXPERIENCE

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### Utah Children's Justice Center

*Victim Assistance Mentor*

Provo, UT

June 2021 – December 2022

- Served children survivors of sexual violence and/or abuse by planning and then doing an activity with them for 2-3 hours each week

